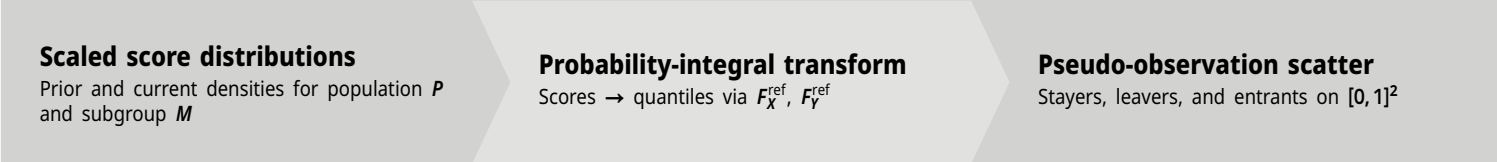


From Scaled Scores to Pseudo-Observations

Stripping Metric Structure via the Probability-Integral Transform

Stayers, leavers, and entrants in copula-based longitudinal inference without longitudinal data



The probability-integral transform. Suppose we observe prior scores $X = (x_1, \dots, x_{n_X})$ and current scores $Y = (y_1, \dots, y_{n_Y})$ for a subgroup S (e.g., a TIMSS country or a NAEP state). The scores live on assessment-specific metric scales whose units carry no intrinsic cross-time meaning. To place both margins on a common, metric-free footing we apply fixed *reference* cumulative distribution functions (CDFs), computed over the full assessment population, to obtain *pseudo-observations* on $[0, 1]$:

$$u_i = F_X^{\text{ref}}(x_i), \quad v_j = F_Y^{\text{ref}}(y_j), \tag{M.1}$$

where F_X^{ref} and F_Y^{ref} are the prior-grade and current-grade population CDFs, respectively. This is the classical *probability-integral transform* (PIT): if F_X^{ref} is the true marginal CDF of X , then $U = F_X^{\text{ref}}(X)$ is uniformly distributed on $[0, 1]$.¹

Population versus subgroup. The reference CDFs are defined over the *full population*, not over S . Consequently, the marginal distribution of U within subgroup S is generally *not* uniform—it reflects S ’s position relative to the population. The outer density strips now encode this directly: the grey background represents the population-level Uniform(0, 1) baseline (density = 1), and the subgroup density curve is plotted relative to that baseline. Where the subgroup curve rises above the grey band, subgroup mass is over-represented relative to the population baseline; where it falls below, it is under-represented. Thin overlay curves show smoothed conditional proportions $P(\text{leaver} \mid u)$ (bottom strip) and $P(\text{entrant} \mid v)$ (left strip).²

Stayers, leavers, and entrants. When prior and current scores come from the *same students* measured longitudinally, we observe linked pairs (x_i, y_i) —and hence pairs (u_i, v_i) —that carry information about how achievement at two time points co-varies within individuals. These *stayers* appear as paired scatter dots in the center panel (population in the grey background cloud; subgroup in the colored foreground). Formally, linked data are draws from a joint law,

$$(U_i, V_i) \stackrel{\text{i.i.d.}}{\sim} C, \quad U_i = F_X^{\text{ref}}(X_i), \quad V_i = F_Y^{\text{ref}}(Y_i), \tag{M.2}$$

where C is the copula encoding within-subject dependence (Sklar’s theorem). *Leavers*—students present in the prior round but not the current one—do not appear in the interior scatter because they are unpaired; they enter the bottom margin through the subgroup- u density and $P(\text{leaver} \mid u)$. *Entrants*—students new to the current round—are analogous on the left margin via the subgroup- v density and $P(\text{entrant} \mid v)$. All three groups contribute to the marginal distributions, but only stayers contribute to the empirical copula.³

Non-linkage and cross-sectional designs. Cross-sectional assessment programs (NAEP, TIMSS) draw independent samples at each occasion. The analyst observes only the marginal distributions,

$$X_1, \dots, X_{n_X} \stackrel{\text{i.i.d.}}{\sim} F_X, \quad Y_1, \dots, Y_{n_Y} \stackrel{\text{i.i.d.}}{\sim} F_Y, \tag{M.3}$$

with no within-subject pairing observed. In terms of the scatter plot, this corresponds to observing only marginal information (CDF strips and density/proportion strips) with no paired interior scatter (copula). The coupling—the copula C that would connect them—is entirely missing and must be supplied by an external model or assumption.⁴

What the marginals preserve and what they discard. The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is monotone and (if F_X^{ref} is continuous) invertible, so no ordinal information is lost. What is discarded is the *metric structure*: the difference $x_2 - x_1$ depends on the scale, but the rank ordering $u_1 < u_2$ is scale-invariant. This is precisely the property exploited by Sklar’s theorem, which factors any joint distribution $H(x, y)$ as

$$H(x, y) = C(F_X(x), F_Y(y)), \tag{M.4}$$

separating the marginal behavior (F_X, F_Y) from the dependence structure (C). The margin strips make this factorization visually explicit: CDF curves are the “bridges” that carry raw scores into copula space, while the density/proportion strips summarize how subgroup composition departs from the population baseline. The interior scatter then reveals the dependence structure that remains after marginal transformation.

¹ The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is invertible when F_X^{ref} is continuous (strictly increasing on its support). In discrete or mixed-continuous settings, the CDF is not strictly monotone and the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ is used. This distinction matters when working with score scales that have mass points (e.g., integer-valued test scores). In such cases, randomized ranks or mid-rank adjustments are sometimes applied to ensure a continuous marginal on $[0, 1]$; see Genest & Nešlehová (2007) and Rüschendorf (2009) for details on pseudo-observation construction under ties.

² The reference CDFs F_X^{ref} and F_Y^{ref} are computed over the *full assessment population*—the complete set of examinees across all subgroups in a given grade and year. They are *not* subgroup-specific. This ensures that all subgroups are mapped to the *same* reference scale, so that a quantile of $u = 0.30$ means “at the 30th percentile of the population” regardless of which subgroup the student belongs to. The subgroup’s marginal density of U then reflects its distributional position within the population. In the margin strips, the grey $U(0, 1)$ band is the population baseline; subgroup density and the conditional leaver/entrant proportion overlays are interpreted relative to it.

³ Stayers are students observed in both the prior and current assessment rounds; after PIT mapping, their linked pairs (u_i, v_i) provide paired draws governed by copula C . Leavers appear only in the prior round (prior-only observations) and entrants appear only in the current round (current-only observations). All three groups contribute to the marginal distributions, but only stayers contribute to the empirical copula.

⁴ In cross-sectional assessment programs (NAEP, TIMSS), independent samples are drawn at each occasion and no student-level pairing exists. The analyst observes only the marginal distributions, so the copula C must be supplied externally—typically from canonical copula tables calibrated on longitudinal datasets in similar content areas and year-spans (see the Overview infographic, Panel B1). The non-linkage premium calculations in the companion appendix quantify the precision cost of this information loss.